

Weather-Based Mosquito Breeding-Favorability Index

Updated recommendations, model scope, disease coverage, validation plan, and legal-risk guardrails

Executive decision

Use the model as a transparent weather-based environmental favorability score for mosquito breeding and operational prioritization. Do not present it as a universal mosquito-borne disease predictor, a case forecast, a clinical tool, or an official public-health warning system.

Document version	v2 - baked-in changes
Suggested model label	Weather-Based Mosquito Breeding-Favorability Index
Formula version	L1-0.2.0
Primary use	City-level screening, comparison, and prioritization
Not for	Disease diagnosis, individual health advice, outbreak prediction, or official alerts without validation and authority review
Prepared date	31 May 2026

Plain-language intended-use statement

Use this exact positioning where possible

This score estimates whether recent temperature, humidity, and rainfall conditions are favorable for mosquito breeding. It is intended to support screening, comparison, and operational prioritization. It does not predict confirmed mosquito abundance, disease transmission, dengue, malaria, chikungunya, Japanese encephalitis, filariasis cases, outbreak probability, or individual health risk.

1. What changes should be baked into the model page?

The current dashboard layout is directionally sound, but the wording should be tightened to reduce scientific and legal exposure. The key changes are naming, normalization, scope, and limitations.

Area	Replace or avoid	Recommended wording
Title	Mosquito-borne disease risk model	Weather-Based Mosquito Breeding-Favorability Index
Scientific claim	Scientifically backed disease prediction	Scientifically motivated, transparent, and locally tunable environmental index
Disease coverage	Covers all mosquito diseases	A weather-only breeding-favorability proxy. Vector-specific calibration is needed for disease-specific use.
Legal posture	Safe / official / proven	For screening and prioritization only; interpret with official surveillance and field observations
Score meaning	Disease risk probability	Relative environmental favorability from 0 to 100

Add this note below the score bands

Dashboard note

Bands are operational categories for comparing environmental favorability across cities. They are not absolute probabilities of mosquito presence, confirmed breeding, disease transmission, or cases.

Recommended level labels

Score	Label	Meaning
0-24	Low	Recent weather is generally unfavorable for breeding.
25-49	Moderate	Some environmental support exists, but conditions are not strongly favorable.
50-74	High	Weather is favorable enough to prioritize field checks and control activities.
75-100	Very High	Weather is strongly favorable; validate with surveillance, larval checks, or operational reports.

2. Recommended v2 formula

Keep the model transparent and configurable. Normalize every component to 0-1 before combining. Temperature remains a limiting multiplier because very cold or very hot conditions can suppress breeding even if moisture is present.

```
score(c,t) = 100 x S_T(T7) x [
    0.30 x S_H(H7)
    + 0.25 x S_R(R0_6)
    + 0.45 x S_R(R7_13)
]

where:
T7    = 7-day mean air temperature
H7    = 7-day mean relative humidity
R0_6  = rainfall total over the most recent 7 days
R7_13 = rainfall total from 7 to 13 days ago
All S_* terms are normalized to the range 0 to 1.
```

Temperature suitability

```
S_T(T) = 0                if T <= 18
        = (T - 18) / 7    if 18 < T < 25
        = 1              if 25 <= T <= 30
        = (35 - T) / 5    if 30 < T < 35
        = 0              if T >= 35
```

This trapezoid is simple and defensible for a lightweight index. Aedes-borne transmission models commonly place broad temperature suitability around 18-34 deg C with peak suitability around 26-29 deg C; the 35 deg C cutoff is deliberately conservative for city-level weather data. [6]

Humidity suitability

```
S_H(H) = clip((H - 40) / 30, 0, 1)
```

This maps relative humidity below 40 percent to low favorability and 70 percent or higher to high favorability. It is a coarse survival/moisture term rather than a species-specific life-table model.

Rainfall suitability

```
S_R(R) = (1 - exp(-R / 30)) x F(R)

F(R) = 1                if R <= 150
      = exp(-(R - 150) / 100) if R > 150
```

This keeps rainfall positive but non-monotonic: some rain creates or refills water-holding habitats, while very heavy rainfall can flush immature stages or remove larvae from some habitats. Aedes eggs and larvae are aquatic/container-linked; CDC summarizes that Aedes eggs develop through larvae and pupae into adults, often on a 7-10 day timescale under favorable conditions. [7]

Why lagged rainfall has the highest weight

Recommended explanation

Lagged rainfall is weighted highest because rain-filled containers and surface water can remain suitable for more than a week, and mosquito development from egg to adult often occurs on an approximately week-scale timeline under favorable conditions. Therefore rainfall from 7-13 days ago can be a useful leading indicator for larvae, pupae, or newly emerging adults.

3. Disease coverage: what the model does and does not cover

The current formula should not be marketed as covering all mosquito-borne diseases. India has multiple mosquito vectors with different breeding ecologies. A single temperature-humidity-rainfall score is best used as a general environmental screening layer unless species-specific data and validation are added.

Disease / group	Primary vector ecology	Fit of current weather-only score	Positioning
Dengue	<i>Aedes aegypti</i> / <i>Aedes albopictus</i> ; container breeding in stagnant water around houses	Reasonable first-pass fit	Can be described as <i>Aedes</i> -relevant breeding favorability, not case prediction. [2]
Chikungunya	<i>Aedes</i> mosquitoes; similar urban container-breeding ecology to dengue	Reasonable first-pass fit	Use alongside dengue/ <i>Aedes</i> language. [1]
Malaria	<i>Anopheles</i> ; <i>An. culicifacies</i> breeds in rainwater pools, puddles, irrigation channels, rice fields, wells, pond margins, and similar habitats	Partial only	Needs an <i>Anopheles</i> module and malaria-specific validation. [3]
Japanese encephalitis	<i>Culex tritaeniorhynchus</i> group; rice-field breeding and animal/bird amplification context	Weak as a standalone model	Needs <i>Culex</i> /rice-field/host context. [4]
Lymphatic filariasis	<i>Culex quinquefasciatus</i> ; polluted water, sewage, drains, cess pits, septic tanks	Weak as a standalone model	Needs polluted-water/sanitation indicators. [5]

Recommended vector-module architecture for broader coverage

To cover the top mosquito-related diseases in India more defensibly, use separate vector modules and combine them only after calibration.

$$\begin{aligned} \text{overall_MBDI}(c, t) = & \lambda_A \times I_{\text{Aedes}}(c, t) \\ & + \lambda_N \times I_{\text{Anopheles}}(c, t) \\ & + \lambda_C \times I_{\text{Culex}}(c, t) \end{aligned}$$

$$\lambda_A + \lambda_N + \lambda_C = 1$$

Disease-specific risk should be a separate layer:

$$\begin{aligned} \text{disease_risk}(d, c, t) = & \text{vector_favorability}(v, c, t) \\ & \times \text{pathogen_presence}(d, c, t) \\ & \times \text{human_exposure}(c, t) \\ & \times \text{local_vulnerability}(c, t) \end{aligned}$$

Product rule

Until the vector modules and disease layers are validated, the product should report environmental breeding favorability, not disease risk or outbreak probability.

4. Published model family to cite

The best published analogue is not a daily weather dashboard; it is the GIS, remote-sensing, weighted-overlay, and AHP family of mosquito-borne disease risk-zone models. Your model is a lightweight operational version of that idea, replacing satellite environmental layers with recent weather inputs.

published weighted-overlay form:

$$MBDI(\text{area}) = 100 \times \sum_k [w_k \times S_k(\text{area})]$$

where S_k is a normalized environmental suitability layer
 and the weights w_k sum to 1.

Reference	Why it matters	How to cite in product docs
Palo et al., GeoHealth, 2025	Remote sensing and GIS study for mosquito-borne disease risk zones in Cuttack, Odisha. Uses environmental layers such as NDWI, NDMI, NDVI, and land surface temperature to classify risk zones. [8]	Use as the Indian all-MBD environmental risk-zone precedent.
Ali & Ahmad, Spatial Information Research, 2019	Kolkata Municipal Corporation study using GIS with AHP weighted overlay and pairwise comparison weighting. [9]	Use as the transparent weighted-overlay / AHP methodology precedent.
Kumar & Agrawal, Nanotechnology for Environmental Engineering, 2020	Gorakhpur case study using Landsat imagery to identify waterlogged areas and mosquito-vector habitat risk zones. [10]	Use as a supporting Indian GIS/remote-sensing vector habitat reference.

Recommended methodology paragraph

Use in report or methodology page

We use a transparent weighted environmental favorability index inspired by published GIS/AHP and remote-sensing mosquito-borne disease risk-zone models in India. Published studies have used reclassified environmental layers such as water presence, moisture, vegetation, and temperature to map mosquito-borne disease risk zones. Our operational version uses recent weather variables - temperature, humidity, and rainfall - as a lightweight city-level proxy for daily monitoring. The output is a relative environmental breeding-favorability score, not a validated epidemiological forecast.

5. Legal and reputational risk controls

The formula alone does not make the product legally safe. Risk is driven by intended use, marketing language, audience, validation status, data practices, and whether users could reasonably treat the score as medical or official public-health guidance. This section is for internal risk reduction and should be reviewed by counsel before public launch.

Risk area	Risk trigger	Control
Overclaiming	Calling the model a disease predictor, outbreak predictor, or official warning system	Use 'environmental breeding favorability' and state that it is for screening and prioritization only.
Medical software	Marketing it as software that detects, diagnoses, monitors, or supports treatment of human disease	Avoid clinical, diagnostic, treatment, individual health, and disease-case prediction claims. Medical-device rules define active diagnostic devices as devices used to supply information for detecting, diagnosing, monitoring, or supporting treatment of health conditions. [11]
Misleading advertising	Claims such as 'scientifically proven to predict dengue' or 'accurate disease risk score'	Substantiate claims, add limitations, and avoid public marketing copy that implies validated disease prediction. The Consumer Protection Act includes powers against false or misleading advertisements. [12]
Privacy	Collecting household locations, reports, symptoms, photos, phone numbers, or identifiable health data	Keep inputs aggregated where possible. If processing personal data, apply DPDP Act controls for consent/notice/purpose/security as applicable. [13]
Scientific pushback	Local entomologists point out vector differences, microhabitats, interventions, or missing field data	Acknowledge that the score is an uncalibrated or pilot environmental proxy unless validated against local surveillance. Maintain versioning and calibration records.

Safe versus unsafe copy

Avoid	Use instead
Predicts dengue risk across India	Estimates weather-based mosquito breeding favorability for city-level screening
Covers all mosquito-borne diseases	Provides a weather-only environmental proxy; vector-specific modules are required for disease-specific coverage
Scientifically proven model	Scientifically motivated, transparent, configurable, and requiring local validation
Official public-health alert	Operational prioritization signal to be interpreted with official surveillance
High score means disease outbreak	High score means recent weather is favorable for breeding; confirm using local field data

6. Required disclaimer and public-facing scope language

Add this disclaimer to the PDF, dashboard methodology page, and any customer-facing methodology documentation. Use a shorter version on small UI surfaces, but keep the complete version available from the dashboard.

Recommended full disclaimer

This score is a weather-based environmental mosquito breeding-favorability index. It estimates whether recent temperature, humidity, and rainfall conditions are favorable for mosquito breeding. It does not predict confirmed mosquito abundance, disease transmission, dengue, malaria, chikungunya, Japanese encephalitis, lymphatic filariasis cases, outbreak probability, or individual health risk. The score is intended for screening, comparison, and operational prioritization only. It should be interpreted together with official surveillance data, local vector-control observations, field inspections, and guidance from public-health authorities.

Short UI disclaimer

Use under chart or score card

Weather-only breeding-favorability score. Not a disease forecast, case prediction, official alert, or individual health-risk estimate.

Recommended methodology card text

This is a transparent weighted formula - deliberately not machine learning - so it can be explained, audited, and tuned through configuration. For each city, the score is:

```
score = 100 x temperature_suitability x (  
    0.30 x humidity_suitability  
+ 0.25 x recent_rain_suitability  
+ 0.45 x lagged_rain_suitability  
)
```

Temperature is a limiting multiplier. If conditions are too cold or too hot, the score is gated toward zero regardless of moisture. Within the suitable temperature band, the score reflects recent humidity and rainfall-derived water availability. Lagged rainfall is weighted highest because rain-filled habitats can remain suitable for more than a week and immature mosquitoes can develop on a week-scale timeline under favorable conditions.

7. How to respond if there is scientific pushback

The correct defense is not 'the model is correct.' The correct defense is that the model is a transparent, scoped screening index with declared assumptions and a validation pathway.

Pushback	Recommended response
This is not a true disease-risk model.	Agree. It is not a disease-risk model. It is an environmental breeding-favorability index and should be combined with surveillance and case data for disease-risk assessment.
Different mosquito species need different habitats.	Agree. The initial score is a weather-only screening layer. The roadmap separates Aedes, Anopheles, and Culex modules for disease-specific use.
Rainfall effects are not linear.	Agree. The v2 rainfall function saturates and includes a heavy-rain penalty to avoid assuming that more rain is always more favorable.
Local water storage, drains, sanitation, and vector-control work are missing.	Agree. These are excluded from the weather-only model and should be added as local layers or accounted for during validation.
The thresholds are arbitrary.	The bands are operational defaults. They must be calibrated against local larval indices, trap counts, complaint data, or surveillance outcomes before high-stakes use.

Validation and governance checklist

- Record formula version, weights, thresholds, and all data sources.
- Back-test against historical larval indices, Breteau Index, Container Index, trap counts, complaint reports, or confirmed public-health surveillance data where available.
- Compare performance by season, geography, and city type rather than relying on a single national accuracy claim.
- Report uncertainty and false positive / false negative examples during pilot use.
- Review wording with public-health experts and legal counsel before public or commercial deployment.
- Keep a change log explaining why any formula, threshold, or weight changed.

Minimum validation claim

Until validation is completed, use: 'pilot environmental indicator' or 'unvalidated weather-only screening score.' After back-testing, update the claim only to the level supported by evidence.

8. Implementation checklist for engineering and product

Item	Recommendation
Inputs	Daily city-level mean temperature, mean relative humidity, daily rainfall. Store source and timestamp.
Rolling windows	Use 7-day temperature and humidity means; split rainfall into recent 0-6 day and lagged 7-13 day windows.
Normalization	Convert each input to 0-1 suitability before applying weights.
Formula	Temperature multiplier x weighted humidity/rainfall mixture. Keep coefficients in config.
Output	0-100 relative environmental favorability score with bands Low, Moderate, High, Very High.
Versioning	Expose model version in the dashboard, for example L1-0.2.0. Log changes to thresholds and weights.
Evidence status	Label as 'scientifically motivated' and 'published-model inspired', not 'proven disease predictor.'
Public copy	Use the short disclaimer next to scores and full disclaimer in methodology docs.

Recommended product release states

Stage	Permitted wording	Do not say
Internal alpha	Experimental weather-only environmental index	Validated risk model
Pilot with partners	Pilot prioritization signal to be compared with field data	Predicts outbreaks
Validated deployment	Locally calibrated environmental indicator with documented performance	Accurate for all cities or all mosquito diseases

Appendix A. Reference notes

These references support the scientific framing, disease-vector distinctions, published model family, and legal-risk guardrails. They do not validate the exact formula; local validation is still required.

- [1] National Center for Vector Borne Diseases Control (NCVBDC), Ministry of Health & Family Welfare, Government of India. Guidelines page for malaria, lymphatic filariasis, dengue, chikungunya, Japanese encephalitis, and related vector-control guidance. Accessed 31 May 2026.
- [2] NCVBDC Dengue FAQ PDF. States that dengue is transmitted by female adult Aedes mosquitoes and that Aedes breed in stagnant water within and around houses, including containers where water stagnates for more than a week. Accessed 31 May 2026.
- [3] NCVBDC Vectors of Malaria. States that Anopheles culicifacies is a main vector of malaria in India and lists breeding places such as rainwater pools, puddles, irrigation channels, rice fields, wells, pond margins, and sluggish streams. Accessed 31 May 2026.
- [4] NCVBDC Question and Answers About Japanese Encephalitis. States that JE is transmitted by rice-field breeding mosquitoes, primarily the Culex tritaeniorhynchus group. Accessed 31 May 2026.
- [5] NCVBDC Filaria Vectors. States that lymphatic filariasis transmission in mainland India involves Culex quinquefasciatus, which prefers polluted waters such as sewage, drains, cess pools, cess pits, and septic tanks. Accessed 31 May 2026.
- [6] Johnson, L.R. et al. (2017). Detecting the impact of temperature on transmission of Zika, dengue, and chikungunya using mechanistic models. PLOS Neglected Tropical Diseases, 11(4), e0005568. DOI: 10.1371/journal.pntd.0005568.
- [7] Centers for Disease Control and Prevention (CDC). Aedes mosquito life cycle materials. CDC states that Aedes eggs, larvae, pupae, and adults form a four-stage life cycle and that egg-to-adult development is often about 7-10 days under favorable conditions. Accessed 31 May 2026.
- [8] Palo, S.K. et al. (2025). Remote sensing and GIS-based study to predict risk zones for mosquito-borne diseases in Cuttack District, Odisha, India. GeoHealth, 9, e2023GH001007. DOI: 10.1029/2023GH001007.
- [9] Ali, S.A. and Ahmad, A. (2019). Mapping of mosquito-borne diseases in Kolkata Municipal Corporation using GIS and AHP based decision making approach. Spatial Information Research. DOI: 10.1007/s41324-019-00242-8.
- [10] Kumar, S. and Agrawal, S. (2020). Prevention of vector-borne disease by the identification and risk assessment of mosquito vector habitats using GIS and remote sensing: a case study of Gorakhpur, India. Nanotechnology for Environmental Engineering, 5. DOI: 10.1007/s41204-020-00084-y.
- [11] Ministry of Health and Family Welfare / CDSCO. Medical Devices Rules, 2017. Definition of active diagnostic medical device includes devices used to supply information for detecting, diagnosing or monitoring, or to provide support in treatment of health conditions. Accessed 31 May 2026.
- [12] The Consumer Protection Act, 2019, India Code. Section on powers against false or misleading advertisements. Accessed 31 May 2026.
- [13] The Digital Personal Data Protection Act, 2023, India Code. Defines personal data and regulates processing of digital personal data. Accessed 31 May 2026.

Appendix B. Final recommended one-line description

One-line product definition

A transparent, weather-based environmental mosquito breeding-favorability index designed for city-level screening and operational prioritization, not disease prediction, individual health-risk estimation, or official public-health warning.